Robotic Arm Control

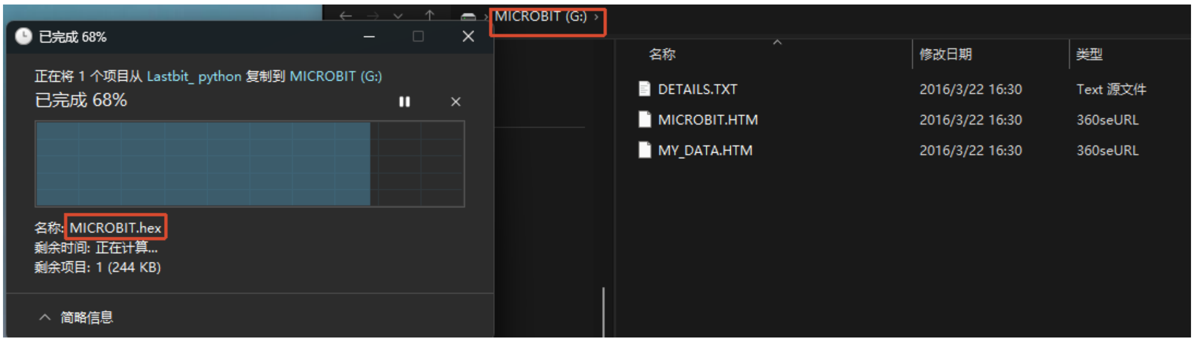
Function Description

The Lastbit is equipped with a three-axis robotic arm. In this section, we will control the three joints of the robotic arm.

S1	**Robotic Arm Gripper Joint**
S2	**Robotic Arm Vertical Joint**
S3	**Robotic Arm Horizontal Joint**

Experiment Steps

Before flashing, make sure `LASTBIT.hex` has been flashed, otherwise an error will occur and the module cannot be imported.



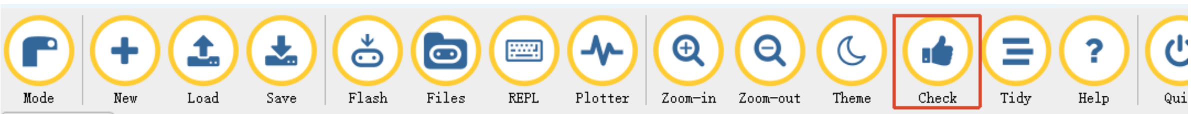
1. Before flashing, toggle the switch to the OFF position, and connect the computer to the micro:bit main board with a microUSB data cable. **Note: use the Microbit interface, not the Charging port of the expansion board.**
2. Open the Mu software and directly copy the program source code we provide. The operation steps are as follows. You can also enter the code yourself in the editor window. Note! All letters and symbols should be entered in English input mode, use the Tab key for indentation, and the last line should end with a blank program.

   Click the `New` button in the toolbar at the top left corner, and a file titled untitled will appear.

3. Copy the content of the `Code Implementation` section provided below into the current file. You will notice a small red dot after the title bar, indicating that the code content has been modified but is in an unsaved state.

Whether you save or not does not affect code checking and flashing.

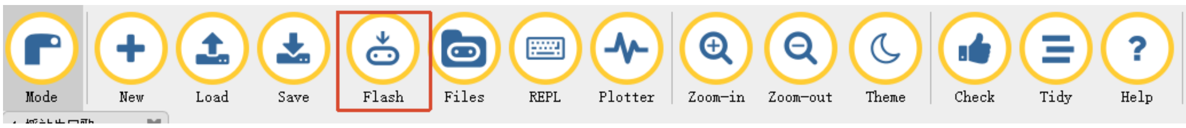
4. At this point, you can click `Check` to check whether there are any problems with the code format.



5. At this point, if you have already connected the MICROBIT to the PC computer following the steps above and flashed `LASTBIT.hex`, the next step is to download the current program to the Microbit.

Click the `Flash` tool in the toolbar to start flashing!

While flashing, the orange-yellow indicator light near the micro:bit main board will blink frequently. When the editor prompts `Copied code onto micro:bit.`, the flashing is complete, and the indicator light will stop blinking afterwards.



6. Unplug the Microusb data cable and toggle the switch to the ON position.

Code Implementation

```
# Robotic Arm Control

from microbit import *
from lastbit import LASTBIT

Lastbit = LASTBIT()
servo_dir = 1
servo_angle = 90
Lastbit.stop_motor()
# Initialize servo angle
```

```
Lastbit.set_all_servos(90)
sleep(1000)

while True:
    # Set all servo angles
    Lastbit.set_all_servos(servo_angle)
    # Change servo angle
    servo_angle += 10 * servo_dir
    # Change servo direction when reaching a certain angle
    if servo_angle >= 160 or servo_angle <= 60:
        servo_dir *= -1

    sleep(50)
```

set_all_servos(angle): controls all servos to move to the `angle` angle.

Experiment Result

After flashing, turn on the switch (toggle the switch to the ON position). The robotic arm will first go to the initial state, then all three joints of the car's robotic arm will move: first it moves to the upper left corner and closes the gripper, then it slowly moves to the lower right corner and opens the gripper, cycling continuously, looking as if it is dancing.

Notes

Make sure the car battery is sufficiently charged, otherwise there may be problems such as inability to control or insufficient torque.